

Example_Scribe



L22

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Generate a Lecture Scribe for exam reference from the given PPT "L22" using the structure from the PDF

"Example_Scribe".

Follow the below two sets of instructions properly to generate the Lecture Scribe :

Instructions : 1) The lecture scribe is to be generated as per the structure given in Example_Scribe 2) Use the content from L22 PPT only and do not add new content

Instructions for following the example scribe : 1) For List of topics covered, the lecture scribe must cover the list of all topics covered in the PPT L22 2) For Explanation of Topics Covered, the lecture scribe must cover detailed explanation of topics covered in PPT 3) For List of Definitions and Theorems, the lecture scribe must include detailed definitions with notations/assumptions along with exact statement of definitions and theorems, detailed proof of theorem step by step (proof only if given in ppt) 4) For Important Examples, try to cover more number of examples from the PPT attached (L22) in such a way that it covers all the concepts discussed. Write entire question for the example, as given in L22 PPT and write detailed step by step solution to examples. INCLUDE EXAMPLES FROM PPT ONLY 5) For the List of Important Formulas, write the list of all the important formulas with reference to topics covered in PPT (as per example_scribe format)



Lecture Scribe: Markov and Chebyshev Inequalities

1) List of Topics Covered

- **Markov's Inequality Recap:** Application to income distribution problems.
- **Chebyshev's Inequality Key Idea:** Utilizing mean and variance to control deviations.
- **Intuition and Geometry:** Understanding tail regions and large deviations from the mean.
- **Proof of Chebyshev's Inequality:** Step-by-step derivation using Markov's Inequality.
- **Comparative Analysis:** Chebyshev vs. Markov bounds in a coin-flipping scenario.
- **Real-Life Applications:** Estimating student score distributions and grade cutoffs.

2) Explanation of Topics Covered

- **Markov's Inequality:** This inequality provides an upper bound on the probability that a non-negative random variable X exceeds a certain value a . It is primarily a "right-tail" bound and requires only the expected value ($E[X]$).
- **Chebyshev's Inequality:** Unlike Markov's, Chebyshev's inequality uses both the mean (μ) and the variance ($Var(X)$) of a distribution. It controls the probability that a random variable deviates from its mean by more than a specific distance a .
- **Intuition:** Geometrically, Chebyshev's inequality bounds the area in the "tail regions" far from the mean. As the spread (variance) around the mean decreases, the probability of finding values far from the center also decreases.
- **Key Insight:** One major advantage of Chebyshev's inequality is that its bound often decreases as the sample size n increases (as seen in the coin-flip example), whereas Markov's bound may remain constant. This makes it a significantly tighter and more useful tool for large datasets.

3) List of Definitions and Theorems

1) Definition: Markov's Inequality

Let X be a non-negative random variable ($X \geq 0$) with a known

expected value $E[X]$. For any $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

- **2) Theorem: Chebyshev's Inequality**

Let X be a random variable with mean μ and variance $Var(X)$. For any $a > 0$, the probability that X deviates from μ by at least a is:

$$P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$$

- **3) Proof of Chebyshev's Inequality**

The proof relies on applying Markov's Inequality to the squared deviation of X from its mean:

1. **Identity:** Note that the event $|X - \mu| \geq a$ is equivalent to the event $(X - \mu)^2 \geq a^2$.
2. **Application:** Define a new non-negative random variable $Y = (X - \mu)^2$. Apply Markov's Inequality to Y with threshold a^2 :

$$P(Y \geq a^2) \leq \frac{E[Y]}{a^2}$$

3. **Substitution:** Since $E[Y] = E[(X - \mu)^2] = Var(X)$, we substitute this into the inequality:

$$P((X - \mu)^2 \geq a^2) \leq \frac{Var(X)}{a^2}$$

4. **Result:** Thus, $P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$.

4) Important Examples

- **Example 1: Income Distribution (Markov)**

Question: In a city, the average annual income is \$40,000. Use Markov's inequality to find an upper bound on the probability that a

randomly selected person earns \$120,000 or more.

Solution:

1. Define $X \geq 0$ as income, with $E[X] = 40,000$.
2. Apply $P(X \geq a) \leq E[X]/a$ with $a = 120,000$.
3. $P(X \geq 120,000) \leq \frac{40,000}{120,000} = \frac{1}{3}$.

Note: Markov cannot bound the probability of earning less than \$10,000 because it only handles the right tail.

- **Example 2: Coin Flipping (Chebyshev vs. Markov)**

Question: Flip a fair coin n times. Let X be the number of heads.

Bound $P(X \geq \frac{3n}{4})$.

Solution:

1. Setup: $X \sim \text{Binomial}(n, 1/2)$, $\mu = n/2$, $\text{Var}(X) = n/4$.
2. Using Chebyshev: $P(X \geq \frac{3n}{4}) = \frac{1}{2}P(|X - \frac{n}{2}| \geq \frac{n}{4})$.
3. Apply bound: $\leq \frac{1}{2} \cdot \frac{n/4}{(n/4)^2} = \frac{2}{n}$.

Insight: The bound $\frac{2}{n}$ decreases as n increases, which is much better than the constant bound provided by Markov's inequality.

- **Example 3: Exam Scores (Real-Life Application)**

Question: Mean exam score is 70 marks and standard deviation is 10.

What fraction of students score below 40 or above 100?

Solution:

1. Identify deviation: The scores 40 and 100 both deviate from the mean (70) by $a = 30$.
2. Apply Chebyshev: $P(|X - 70| \geq 30) \leq \frac{10^2}{30^2} = \frac{100}{900} = \frac{1}{9}$.
3. At most 1/9 of students are far from the mean.

- **Example 4: Exam Grades with Variance (Exercise 5.8b)**

Question: The average grade is 70% and the standard deviation is 5%. The "A" cutoff is 90%. Use Chebyshev's inequality to find an upper bound on the fraction of "A" students.

Solution:

1. Identify parameters: $\mu = 70$, $\sigma = 5 \Rightarrow \text{Var}(X) = 25$.
2. Identify deviation: An "A" student ($X \geq 90$) deviates from the mean by at least 20 ($|X - 70| \geq 20$).
3. Apply bound: $P(X \geq 90) \leq P(|X - 70| \geq 20) \leq \frac{25}{20^2}$.

4. Calculation: $\frac{25}{400} = \frac{1}{16} = 0.0625$ or 6.25%.

5) List of Important Formulas

Concept	Formula	Notes
Markov’s Inequality	$P(X \geq a) \leq \frac{E[X]}{a}$	Requires $X \geq 0$
Chebyshev’s Inequality	$P(X - \mu \geq a) \leq \frac{\text{Var}(X)}{a^2}$	Uses mean and variance
Variance Calculation	$\text{Var}(X) = E[(X - \mu)^2]$	Fundamental to Chebyshev proof
Binomial Mean	$\mu = np$	Used in coin-flip example
Binomial Variance	$\text{Var}(X) = npq$	Used in coin-flip example
		